

Framework for enabling smartphone-based ecological cognitive neuroscience

Task: conjunctive search on smartphones
ERP study with comparison of behaviour and physiology to conventional studies.

Issues to resolve:

1. Timing accuracy of stimulus presentation and response recording.
2. Eye-tracking with selfie camera (+IR?)
3. Synchronization of all data streams

Datasets generated

1. ET-BCI : simultaneous recording of gaze + EEG + action, with calibration/grand truth data
2. Facial video + wearable ET for optimizing video-based ET
3. EEG data of mobile vs computer based task

Publications

1. Methods paper : Describe the setup/framework and compare results from phone vs screen based task (Nature methods level quality)
- ✓ 2. ET-BCI dataset with DLC validation :
Ground truth key location and timing data to be validated against DLC data in task. Then same DLC can be used in natural context.
- ✓ 3. Facial video based eye tracking (gaze + pupillometry)
- ✓ 4. Expression based attention estimation?
- ✓ 5. ET-BCI analysis paper

Things to do:

1. Find appropriate conjunctive search task to compare ERPs
 2. Implement task in javascript.
 3. Task should be designed to enable validation of timing consistency
 4. Implement an app with webview to show the task in full screen
 5. Record all touch events (timing & coordinates)
 6. Record screen
 7. Use wearable ET + sync with EEG
 8. Sync phone events with EEG by streaming taps over USB
 9. Repeat toggles at end to validate sync.
- Tap based screen toggling (see below)

ET-BCI

Task: ET PoV video $\xrightarrow{DLC}$ thumb location
+
tap loc / timing

How to sync? $\updownarrow$



Phone tap data $\rightarrow$ tap loc / timing

tap loc / timing

1. At start of exp, show full black screen
2. Toggle on tap 10 times

timing,
location,
count

Validate DLC accuracy!

Use to sync phone tap & ET PoV

* Publish!

Phone tap data + EEG + gaze $\rightarrow$ analyze

EEG predicts tap timing

Gaze predicts tap location

Task based

DLC tap data + EEG + gaze $\rightarrow$ analyze

How well does he above translate to real time use?

Free scrolling based

* Publish!

Facial video based ET

ET eye data → ground truth
 selfie video → input data } How to sync?



1. ET pov + taps synced as in ET-BCI
2. Start selfie recording on first tap
 Include a button on first blank screen!
3. Ensure low latency betⁿ button tap & recording onset.
 How?
 Attach a small reflector to the face and sync toggles!

Train & test model

* Publish!

Expression based attention estimation

selfie video
 +
 ERP / accuracy } How to sync?
 ↳ already synced to phone tap data.
 ↳ Extract attentiveness metric(s)

Train & test model

* publish!

Methods paper

Timing validation for stimulus & response

- Assume tap timestamps reliable **check!**
i.e. response times are accurate
- Next stimulus fixed duration after response
(or known variable delay $\rightarrow$ better)
- Record screen to obtain actual latency & compare
- How to sync screen recording and stimulus display? $\rightarrow$ Use mutual screen toggles
 - Tap times accurate
 - Match taps to toggle in video
 - Estimate jitter **At least 10 times toggle**
 - Match timestamps

EEG

} How to sync?

Phone tap data

} m1. send taps over LSL?

- m2. Taps synced with ET PoV $\rightarrow$ 30fps
- ET PoV synced with ET **imagination**
- ET synced with EEG **various uncertainty**

Validate LSL

accuracy for applications
without wearable ET

Reduced by using 10 toggles!
Estimate final uncertainty!